Estonian-Finnish Olympiad - 2009

1. Boat ***(9 points)*** Consider a boat of effective mass m, which moves along the (horizontal) x-axes, and for which the friction force of the water is given by $F_f = -\alpha v$, where v is the velocity of the boat. (Effective mass is somewhat larger than the real mass, because any change of the boat's velocity incurs a velocity change of those regions of water, which are close to the boat; hence, the boat's inertia is complemented by the inertia of the surrounding water.)

i) Prove that during this motion,

$$v + kx = \text{Const},$$

and find the factor k.

ii) Now, a boy of mass M moves inside the boat, back and forth, with velocity $u(t)$ (also, along the x-axis). What term must be added to the conservation law $v + kx = \text{Const}$, to describe this situation?

iii) Now, let the boat rest initially close to the coast and ready to depart. The boy jumps from the coast into the boat, turns around and jumps back to the coast. The horizontal component of the boy's velocity during the first jump was u_1, and during the second jump $-u_2$. What is the maximum distance s travelled by the boat? Neglect the effect of a vertical motion on the horizontal friction force of the water.

2. Nanoclock ***(10 points)***

Nanotechnology allows controllable fabrication of very small structures. Let us consider a tiny homogeneously charged thin ring, having a radius R and carrying a positive net charge Q.

i) Find the electric potential φ at a point P, which is on the axes of the ring, at a distance z from the centre of the ring.

ii) Find the electric field E at the point P.

iii) Show that the force acting on an electron moving along the symmetry axes in the vicinity of the centre of the ring ($|z| \ll R$) is harmonic (i.e. depends linearly on z).

iv) Define the frequency of oscillations of such an electron. Use numerical values $R = 1\mu\text{m}$ and $Q = 1.0 \times 10^{-13}\,\text{C}$.

v) Now, let us assume that the electron can perform also off-axis movements. Is the position at the symmetry centre of the ring (on-axis, $z = 0$) stable or unstable? Motivate your answer.

Hint: You may use the approximate formula

$$(1 + x)^\alpha \approx 1 + \alpha x + \tfrac{1}{2}\alpha(\alpha - 1)x^2.$$

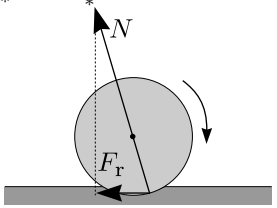
3. Ball ***(8 points)*** A homogeneous ball of radius R and mass m is thrown horizontally onto a table at the moment of time $t = 0$. Its initial velocity before contact is purely horizontal and equal to v; it is non-rotating. Coefficient of kinetic friction between the table and the ball is μ.

i) Find the moment of time t, when the ball stops sliding, i.e. starts rolling without sliding.

ii) Calculate the angular velocity of the ball ω_* and its total mechanical energy E_* at the moment when it stops sliding. In the case of a hollow sphere, would the energy E_* be larger or smaller than in the case of a homogeneous ball?

iii) Now, assume that the horizontal surface is treated so that the coefficient of kinetic friction depends on the horizontal coordinate x as $\mu = a + b\cos x$ (with $a > b$). Find the expression for the terminal mechanical energy E'_* in this new case.

iv) Now, let us return to the case of constant kinetic friction coefficient μ. However, let us assume that the surface is not perfectly rigid (e.g. covered by a felt cloth). This gives rise to a second friction force — the rolling friction force $F_r = \mu_r mg$. Unlike the kinetic friction force, it is not tangential to the touching point of the ball and the surface. Instead, it can be interpreted as the horizontal component of the surface reaction force (the entire reaction force is, of course, normal to the surface), see Fig. Find the expression for the terminal mechanical energy E''_* in this case. What is the most important (qualitative) difference between the expressions for E_* and E''_*?



Hint: The moment of inertia of a ball is $I = \frac{2}{5}MR^2$.

4. Black box ***(9 points)*** *Equipment:* electrical black box with three outlets, battery, voltmeter.

It is known that inside the black box, there are three resistors (connected with wires in an unknown manner), the smallest of which is $R_1 = 100\,\Omega$. Find the value of the largest resistor R_3. What can be said about the middle-valued resistor R_2? Estimate the uncertainties of your results.

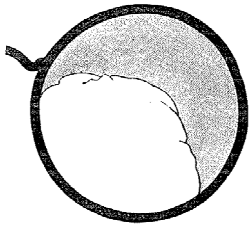
5. Pencil ***(6 points)*** *Equipment:* pencil, paper, ruler.

Determine the coefficient of friction of the pencil's graphite core against the paper. Estimate the uncertainty.

6. Spring ***(7 points)*** *Equipment:* helical spring of known mass $m = 19 \pm 0.5\,\mathrm{g}$, measuring tape, a load of unknown mass.

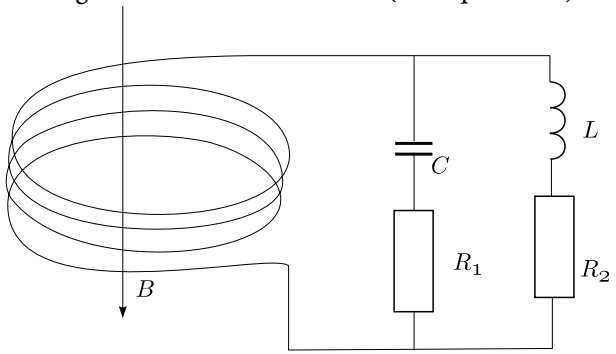
Determine the mass of the load. Estimate the uncertainty.

7. Soap film ***(6 points)*** Lord Rayleigh had in 1891 a lecture about taking photos of physical processes. Among others, he showed a photo of a soap film, which is falling apart (see Fig.). Instead of a flash, he used an electric spark (well, nowadays the flashes are also based on electric sparks). Estimate, how precise must have been the timing, i.e. estimate the time for a soap film to fall apart. Let the thickness of the soap film be $h = 1$ μm, the ring diameter $D = 10\,\mathrm{cm}$ and the surface tension $\sigma = 0.025\,\mathrm{N/m}$.



Hint: you may use a model, according to which the already broken part of the soap film gathers into a single front and moves all together towards the still preserved part of the film.

8. Magnetic pulse ***(7 points)*** Consider an electric circuit consisting of a coil of negligibly small inductance, consisting of $N = 10$ turns and with the surface area of a single loop $S = 10\,\mathrm{cm}^2$), resistors $R_1 = R_2 = 3\,\Omega$, capacitor $C = 0.2\,\mathrm{F}$, and an inductance $L = 1H$, connected as shown in Fig. At the moment of time $t = 0$, a magnetic field, parallel to the axis of the coil is switched on. The inductance of the magnetic field starts growing linearly, starting from $B = 0$ until the maximal value $B = 1\,\mathrm{T}$ is achieved at $t = 10\,\mathrm{ms}$. Further, the inductance of the magnetic field remains constant (and equal to 1T).



i) Find the current through the resistors R_1 and R_2 at the moment of time $t_1 = 5\,\mathrm{ms}$.

ii) Find the current through the resistors R_1 and R_2 at the moment of time $t_2 = 15\,\mathrm{ms}$.

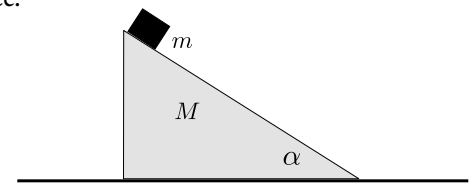
iii) What is the net charge passing through the resistor R_2?

9. Stratostat ***(5 points)***

i) Show that the pressure of an isothermal gas of molar mass μ follows the law $p = p_0 e^{-\alpha z}$, where p_0 is the pressure at the origin, and z is the height. Find the constant α. The temperature is T, the free fall acceleration is g.

ii) Consider a stratostat, the envelope of which (a freely deformable non-elastic sack) is filled at the Earth's surface by helium to the volume fraction of $\beta = 10\%$. At which height h does the helium expand so that it fills the entire volume of the stratostat? The molar masses of the air and helium are $\mu_a = 29\,\mathrm{g/mol}$ and $\mu_{He} = 4\,\mathrm{g/mol}$, respectively. You may neglect the temperature variations of the atmosphere, and use the value $T = 250\,\mathrm{K}$.

10. Wedge ***(5 points)*** A wedge of mass M is kept at rest on an horizontal surface, and a block of mass m is kept on the wedge at the height h from the surface. The angle of the wedge is α, see Fig. There is no friction neither between the block and wedge nor between the surface and the wedge. The system is released into a free motion. Find the time t needed for the block to reach the surface.



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1. Boat ***(9 points)***

i) From the Newton II law, $m\frac{dv}{dt} + \alpha\frac{dx}{dt} = 0$. Multiplying this equation by dt, we obtain $dv + \frac{\alpha}{m}dx = 0$. Integrating (i.e. summing over all the small increments dx and dv) this equation leads us to

$$v + \frac{\alpha}{m}x = \text{Const},$$

i.e. $k = \frac{\alpha}{m}$.

ii) We proceed in the same way as before, but we need to add the interaction force between the boat and the boy: $m\frac{dv}{dt} + \alpha\frac{dx}{dt} = M\frac{du}{dt}$. [Note that since the right-hand-side of this equation is the interaction force, it goes to zero, if the boy leaves the boat. Therefore, if we want to keep this equation correct even after the boy leaves the boat at the moment of time $t = t_*$, we must assume $u(t) \equiv u(t_*)$ for $t > t_*$.] Similarly to the previous section, we obtain

$$v + \frac{M}{m}u(t) + \frac{\alpha}{m}x = \text{Const}.$$

iii) We use the conservation law of the previous section, and compare the value of the left-hand-side immediately before the boy lands into the boat with its value after a very long time. Bearing in mind that we need to substitute $u(t \to \infty) = -u_2$ (see above), we obtain

$$0 + \frac{M}{m}u_1 + 0 = 0 - \frac{M}{m}u_2 + \frac{\alpha}{m}s.$$

So,

$$s = \frac{M}{\alpha}(u_1 + u_2),$$

i.e. the result is independent of how long time did the boy spend in the boat.

2. Nanoclock ***(10 points)***

i) All the charges of the ring are at the same distance from the point P, $l = \sqrt{R^2 + z^2}$. So, according to the superposition principle, the potential is the sum of potential of all the charges,

$$\varphi = kQ/\sqrt{R^2 + z^2}.$$

ii) $E = -\frac{d\varphi}{dz} = kQz/(R^2 + z^2)^{3/2}$.

iii) For $|z| \ll R$ we may approximate $(R^2 + z^2)^{3/2} \approx R^3$, so that $E = \approx kQz/R^3$.

iv) From the Newton II law, $m\ddot{z} = -ekQz/R^3$, i.e. $\omega^2 = ekQ/mR^3$ and $f = \frac{1}{2\pi R}\sqrt{\frac{ekQ}{mR}} \approx 5.6 \times 10^{12}$ Hz.

v) At the origin, there are no charges; hence, the electric field lines cannot neither start nor end there. Consider a tiny coaxial cylinder embracing the origin. The field lines exit the cylinder through its bottom and top surfaces (because there is an electric field $E = \approx kQz/R^3$). Hence there must be field lines entering the cylinder through its side surface. This implies a radial repelling force for an electron situated at that surface, i.e. instability.

Remark: In the same way, one can prove a theorem, electrostatic equlibria are always unstable.

3. Ball ***(8 points)*** A straightforward way to solve parts **i)**, **ii)**, and **iii)** is to use conservation of angular momentum with respect to any axis laying on the surface (there is no torque with respect to these axes). Angular momentum at the beginning: $L_0 = mvr$. Rolling ball is always rotating around the touching point with the ground (although the location of that point is constantly changing). From Steiner's theorem we know that for a rotation axis that is located at the distance r form the center of mass, the moment of inertia is $I' = I + mr^2$. Therefore the angular momentum for rotating ball is $L' = I'\omega$ and clearly $L_0 = L'$, therefore $mvr = (I + mr^2)\omega, \quad \omega = \frac{mvr}{I+mr^2}$.

i) The torque with respect to center of mass: $M = mg\mu r$. Since $M\Delta t = I\Delta\omega$, and the sliding stops when angular speed has reached value ω, we obtain the corresponding time: $t = \frac{I\omega}{mg\mu r} = \frac{Iv}{g\mu(I+mr^2)}$.

ii) Clearly, $\omega_* = \frac{mvr}{I+mr^2}$ and $E_* = \frac{1}{2}(mv_*^2 + I\omega_*^2) = \frac{1}{2}(mr^2\omega_*^2 + I\omega_*^2) = \frac{m^2v^2r^2}{2(I+mr^2)}$. If I increases then E_* decreases.

iii) Since we didn't make any assumptions about the functional form of μ while deriving w_*, the results w_* and E_* of **ii)** are still valid.

iv) The net force $F = F_\mu + F_r$ causes the decrease of translational velocity: $m\Delta v = F\Delta t, \quad v'(t) = v - (\mu + \mu_r)gt$. The sliding stops when $\omega(t) = v'(t)/r$. Note that the torque with respect to center of mass is only caused by kinetic frictional force, therefore $\omega(t) = \frac{mg\mu r}{I}t$. We get an equation for the time of the termination of sliding: $\frac{mg\mu r^2}{I}t = v - (\mu + \mu_r)gt, \quad t = \frac{Iv}{g\mu(I+mr^2)+g\mu_r I}$. The corresponding angular speed ${\omega_*}'' = \frac{mvr}{I+mr^2+\frac{\mu_r}{\mu}I}$, and energy ${E_*}'' = \frac{1}{2}(I + mr^2){\omega_*}^2$. Clearly, ${E_*}'' < E_*$.

4. Black box ***(9 points)*** Independetly of whether there is a triangular or a star connection, one can measure the ratio of the resistances a pair of resistors by connecting the battery to two outlets — let these be the outlets 1 and 3, and measuring the voltages between the outlets 1 and 2, and between the outlets 2 and 3. For a star connection,

$$\frac{R_3}{R_1} = \frac{V_{23}}{V_{12}},$$

where R_i denotes the resistor closest to the i-th outlet; for a triangular connection,

$$\frac{R_3}{R_1} = \frac{V_{12}}{V_{23}},$$

where R_i denotes the resistor farthest away from the i-th outlet; for a triangular connection,

In such a way we find

$$\frac{R_3}{R_1} \approx 4{,}7, \quad \frac{R_3}{R_2} \approx 3{,}3, \text{ and } \quad \frac{R_2}{R_1} \approx 1{,}4,$$

implying $R_1 = 100\,\Omega$, $R_2 = 140\,\Omega$ and $R_3 = 470\,\Omega$; or

$$\frac{R_1}{R_3} \approx 4{,}7, \quad \frac{R_2}{R_3} \approx 3{,}3, \text{ and } \quad \frac{R_1}{R_2} \approx 1{,}4,$$

$R_3 = 100\,\Omega$, $R_2 = 330\,\Omega$ and $R_1 = 470\,\Omega$. So, we can conclude that the largest resistance is $470\,\Omega$, and the middle-valued resistance is either $330\,\Omega$ or $140\,\Omega$.

5. Pencil *(6 points)* We put the pencil with its graphite end against a horizontal sheet of paper, and push the other end with a fingertip (without holding between the fingers and thereby possibly giving a torque to it). Then, there will be a purely longitudinal stress inside the pencil (there is no bending of the pencil). If we push strongly enough, we may neglect the weight of the pancil in the balance of torque with respect to the fingertip. So, equilibrium implies that the resultant force of the friction and reaction forces at the graphite end are along the axis of the pencil. This is possible, if $\tan\alpha \le \mu$, where α is the angle between the pencil and a vertical line. So, we increase α wile pushing the pencil, and determine the position α_0, when the pencil starts sliding; $\tan\alpha_0$ can be calculated as $\tan\alpha_0 = x/\sqrt{l^2-x^2}$, where l is the length of the pencil, and x is the length of the projection of the pencil onto a horizontal plane.

6. Spring *(7 points)* First, we measure the lengthening of the spring under its own weight. The relative lengthening of each loop is propotional to the order number of the loop. So, the average value of those legth increments is half of the largest increment (for the topmost loop). The topmost loop is deformed by the weight of the whole spring, hence the average deformation corresponds to the half-weight of the spring,

$$kx_1 = mg/2.$$

Now we add a load to the lower end of the spring and measure the new deformation x_2. Since the additional weight of the load increases the stress of all the loops by the same value Mg, each loop is deformed additionally by the same length increment. The sum of those additional increments is $x_2 - x_1$; according to the Hook's law, $k(x_2 - x_1) = Mg$. So, $\frac{x_2}{x_1} - 1 = \frac{2M}{m}$, and

$$M = \frac{m}{2}\frac{x_2 - x_1}{x_1} \approx 6\,\mathrm{g}.$$

7. Soap film *(6 points)* Suppose the area of a broken part of the soap film is S. The corresponding surface energy is $E_S = 2\sigma S$. The mass of the soap water that was previously located at the broken part is $m = Sh\rho$, where $\rho = 10^3 kg/m^3$. Assuming that the surface energy is transformed to the kinetic energy of the moving front, we get an equation for the speed of the front: $2\sigma S = \frac{1}{2}mv^2, \quad 4\sigma S = Sh\rho v^2$. Therefore $v = \sqrt{\frac{4\sigma}{h\rho}} = \sqrt{\frac{4\cdot 0.025}{10^{-6}\cdot 10^3}} m/s = 10 m/s$ and we can estimate the time of breakup of the soap film to be $t = D/v = \frac{1}{100}s$.

8. Magnetic pulse *(7 points)* Since we can neglect the inductance of the coil, it performes as a voltage source, which outputs $U = NSB/\tau = 1\,\mathrm{V}$ during the time period between $t = 0\,\mathrm{ms}$ and $10\,\mathrm{ms}$, and $0\,\mathrm{V}$ otherwise. The characteristic time scales of the RC and LC cirquits are $\tau_1 = R_1 C = 0.6\,\mathrm{s}$ and $\tau_2 = L/R_2 \approx 0.3\,\mathrm{s}$. So, for both cirquits, the processes are very fast, i.e. the capacitor is effectively short-circuited, and almost all the voltage falls on the inductance.

i) According to the considerations given above, $I_1 = U/R_1 \approx 0.33\,\mathrm{A}$. As for I_2, it starts growing from $0\,\mathrm{A}$ at $t = 0$ at a rate, given by $L\frac{dI}{dt} = U$, i.e. $I_2 = Ut_1/L = 5\,\mathrm{mA}$.

ii) When the voltage U is switched off (at $t = \tau = 10\,\mathrm{ms}$), the capacitor will (almost completely, because $t_2 - \tau \ll R_1 C$) retain the charge it has accumulated, $Q = I_1\tau$. All the voltage of the capacitor (Q/C) will fall on the resistor R_1, so that $I_1' = Q/R_1 C = U\tau/R_1^2 C \approx 5.6\,\mathrm{mA}$. As for the inductance, it will retain (almost completely, because $t_2 - \tau \ll L/R_2$) the current it has acquired during the first $10\,\mathrm{ms}$, $I_2' = U\tau/L = 10\,\mathrm{mA}$.

iii) Since the current in R_2 will decay very slowly, as compared to its growth during the first $10\,\mathrm{ms}$, we can neglect the charge passing thorugh it during $t < \tau$. Then we can write the Kirhoff's law in the form $L\frac{dI}{dt} + R_2\frac{dq}{dt} = 0$, from where $LdI + R_2 dq = 0$, and $L\Delta I = -R_2\Delta q$. Since $\Delta I = -I_2'$, we obtain $\Delta q = LI_2'/R_2 = 3.3\,\mathrm{mC}$.

9. Stratostat *(5 points)*

i) Consider the pressure difference at heights $z + dz$ and z: $dp = -\rho g dz$ (the difference is simply due to the weight of the layer dz). The density can be found using the state equation of ideal gas: $pV = \frac{m}{\mu}RT, \quad \rho = \frac{m}{V} = \frac{\mu p}{RT}$. Therefore $\frac{1}{p}\frac{dp}{dz} = -\frac{\mu g}{RT}$. Note that we have a derivative of a logarithm: $\frac{1}{p}\frac{dp}{dz} = \frac{d\ln p(z)}{dz}$. Therefore $\ln(p/p_0) = -\alpha z, \quad p = p_0 e^{-\alpha z}$, where $\alpha = \frac{\mu g}{RT}$.

Alternatively, we can derive this law from the Boltzmann distribution for the particle density $n = n_0 e^{-U/kT}$, where U is the potential energy of a molecule. Bearing in mind that for a constant temperature, the pressure is propotional to the density, we obtain $p = p_0 e^{-U/kT}$. Substituting $U = mgz = \frac{\mu}{N_A}gz$ and $R = kN_A$, we obtain the same result as above.

ii) Clearly, the pressures inside and outside of the stratostat are equal and depending on the height as $p = p_0 e^{-\alpha z}$, where $\alpha = \frac{\mu_a g}{RT}$. For helium inside the sack $pV = const$. Let the volume of the stratostat be V_S, therefore $p_0\beta V_S = pV_S$, where p is the pressure at the height, where helium has filled the entire volume. Since $p = \beta p_0$, we obtain $\frac{\mu_a g}{RT}h = -\ln\beta, \quad h = \frac{RT}{\mu_a g}\ln\frac{1}{\beta}, \quad h = 17\,\mathrm{km}$.

10. Wedge *(5 points)* The center of mass of the system doesn't move, therefore $Mu = mv_h$, where u and v_h are the horisontal components of the velocities of the wedge and block in the lab frame at some instant of time. In the wedge's frame, the block has horisontal velocity $v' = v_h + u = v_h(1 + \frac{m}{M})$. Since the block is sliding down the wedge, the vertical component of the block has to be $v_\downarrow = v'\tan\alpha$. The total kinetic energy in the lab frame $E_K = \frac{1}{2}(Mu^2 + mv_h^2 + mv_\downarrow^2)$. Substituting the relevant quantities and simplifying, we obtain

$$E_k = \frac{m}{2}v_\downarrow^2\beta, \text{ where } \beta \equiv \frac{M}{M+m}\cot^2\alpha + 1.$$

The kinetic energy equals to the change in potential energy: $\frac{m}{2}\beta v_\downarrow^2 = \Delta E_P = mg(h - h')$. By differentiating and noting that $dh = dt\cdot v_\downarrow$, we get $\beta dv_\downarrow = gdt, \quad t = \frac{\beta}{g}v_{\downarrow\max}$. From the above written energy conservation law we get $v_{\downarrow\max} = \sqrt{\frac{2gh}{\beta}}$; so we finally have $t = \sqrt{\frac{2\beta h}{g}} = \sqrt{\frac{2h}{g}\left(\frac{M}{M+m}\cot^2\alpha + 1\right)}$.